Applied Deep learning (XAI602C)

Deep learning is a branch of Machine Learning that uses layered neural networks to learn patterns from data and make decisions or predictions.

In simple terms

It's a way for computers to learn from examples, similar to how humans learn from experience—but using large amounts of data and computing power.

How it works

Deep learning models are inspired by the human brain and are built using Artificial Neural Networks:

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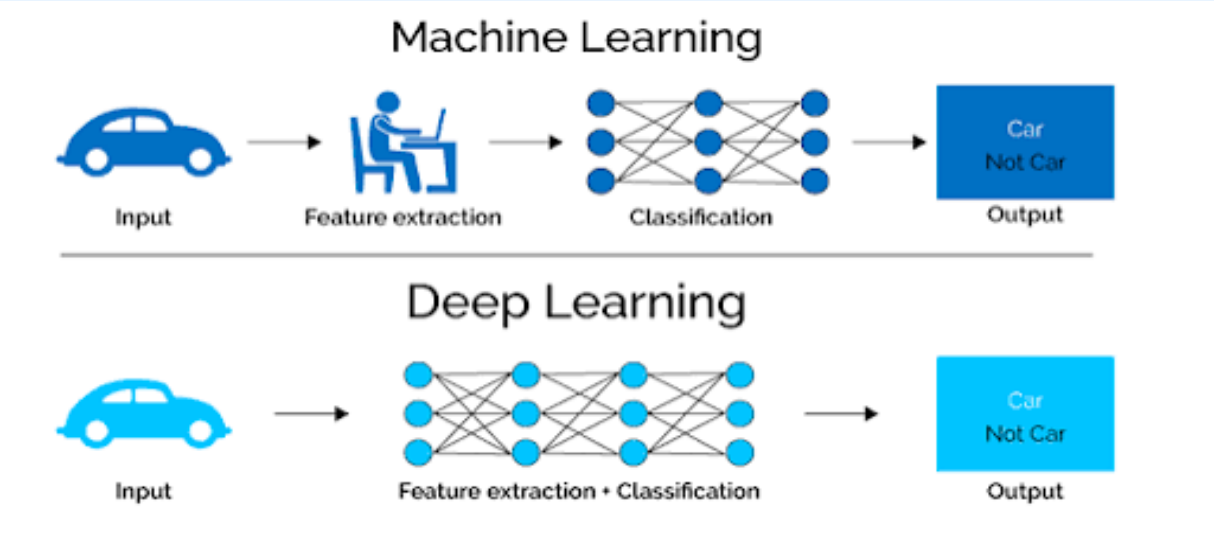
Each layer processes data and passes it to the next
The “deep” in deep learning means many layers
For example:
An image goes in → layers detect edges → shapes → objects → final output (like “cat” or “dog”).
Where it’s used
Deep learning powers many modern technologies:
Voice assistants (like Siri and Google Assistant)
Image recognition (face unlock, medical scans)
Self-driving cars
Language translation and chatbots
Why it’s important
Deep learning is powerful



A neural network is a computer system inspired by how the human brain works. It is a key part of Artificial Intelligence and is widely used in Machine Learning.

Simple definition

A neural network is a group of connected “neurons” (small processing units) that work together to learn patterns from data and make decisions.

Structure of a neural Network

Input Layer → Hidden Layer(s) → Output Layer
(Data) (Processing) (Result)

Neural networks are built using layers:

1. Input Layer – receives data (like numbers, images, text)

2. Hidden Layers – processes